Patient Encounter Cost Predictive Model

University of Virginia
DS5003 - Group 10

Claudio Cela
cc5hu@virginia.edu

Claire Sullivan
kyq5rs@virginia.edu

Jack Hays
jth2uyr@virginia.edu

Sree Prabhav Bandakavi
kqy9hc@virginia.edu

1 Target Audience

Hospital Patients, Public Health Officials, and Healthcare Providers

2 Executive Summary

We developed a predictive model that accurately provides personalized estimates of patient cost burden during hospital encounters to improve healthcare cost transparency and support better patient decision-making.

In the current U.S. healthcare system, many uninsured and underinsured individuals delay or avoid necessary care due to uncertainty about potential expenses, leading to increased health risks and financial strain. The absence of reliable, upfront pricing information forces patients to make critical decisions without a clear understanding of costs. To address this gap, we leverage synthetic patient data from Synthea to build a machine learning–based framework that captures the complex factors influencing healthcare expenses and generates individualized cost predictions. By offering more accurate and accessible cost estimates, this approach aims to reduce financial uncertainty, improve access to care, and empower patients to make more informed healthcare choices.

3 Introduction

Healthcare costs in the United States remain highly variable and often unpredictable, creating significant challenges for patients—particularly those without adequate insurance coverage. Uninsured individuals disproportionately rely on emergency departments for care, often facing substantial and unexpected financial burdens as a result [2]. More broadly, recent evidence suggests that nearly 38% of adults delay or avoid medical care, with cost uncertainty serving as a primary contributing factor [3]. This lack of financial transparency forces patients to make critical healthcare decisions without a clear understanding of potential expenses.

The consequences of delayed care are well documented. Prior research has shown that postponing treatment is associated with worsening health outcomes and increased utilization of more intensive and costly healthcare services [6]. Despite ongoing efforts to improve price transparency, existing tools often fail to provide reliable, personalized estimates that reflect the complexity of individual patient encounters.

A wide range of interacting factors, including patient demographics, underlying conditions, and the nature of the medical encounter itself, influence healthcare costs. This complexity leads to significant variability in costs, even for similar procedures, making accurate prediction inherently challenging. Recent studies have demonstrated that statistical and machine learning approaches can effectively model these nonlinear relationships and improve the accuracy of medical cost prediction [4].

In this study, we develop a predictive framework to estimate a patient's cost burden during hospital encounters using realistic synthetic data generated by Synthea. By leveraging data-driven methods to capture complex patterns in healthcare utilization, our approach aims to provide personalized and actionable cost estimates. Ultimately, this work aims to improve financial transparency for patients.

4 Dataset

4.1 Synthea

The data used for this project is simulated, open-source patient population data downloaded from the Synthea data simulator, produced by The MITRE Corporation. The dataset downloaded was "1K Sample Synthetic Patient Records, CSV," consisting of 1,000 simulated patients and containing information on their conditions, insurance, demographics, healthcare providers, encounter costs, and so on [5].

4.2 Tableau EDA

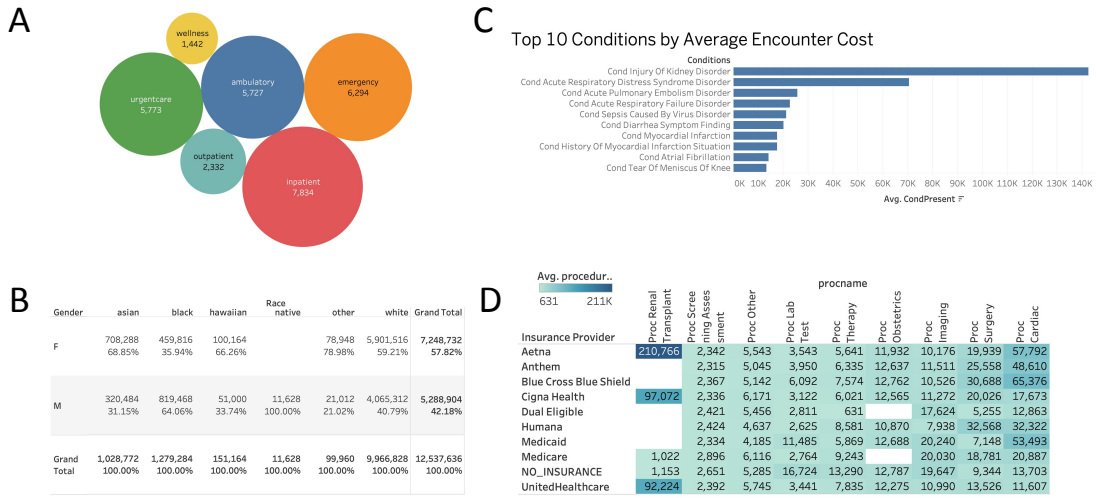

Figure 1: (A)Average encounter cost varies by encounter class, with higher-cost classes highlighted by larger markers. (B)Gender distribution differs across races, with percentages showing within-race composition. (C)A small subset of conditions drives the highest average encounter costs. (D)Procedure costs vary across categories and insurance providers, with higher costs concentrated in specific combinations. The Interactive Dashboard can be accessed at `https://public.tableau.com/views/DS5003_Tableau/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link`

4.3 Pre-processing

The data were pre-processed to yield a resulting data frame merged on encounter ID and containing relevant predictor columns, including age, gender, race, healthcare costs, conditions, medical devices, hospital city, and others. The team decided to treat each hospital encounter as its own observation, rather than each patient, to reflect the instantaneous medical conditions and healthcare costs of a patient over time, rather than grouping the encounters into a patient-level summary. Total encounter cost was calculated as the sum of the base encounter cost and the procedure cost for that encounter. A key decision made in the pre-processing was the choice to one-hot encode all 180 conditions present in the dataset, as we primarily wanted to investigate the role of different conditions on encounter cost. Additionally, due to the large number of unique procedures and medications, categorical bins were used to summarize similar areas, resulting in 8 and 9 categorical bins for procedures and medications, respectively. We performed minor cleaning to get the resulting dataframe into model-ready form, including dropping unique identifying columns and verifying column data types. The resulting dataframe had 61,459 observations and 240 columns. We transformed the encounter cost on a log scale and split the data into a train/test/validation set, preparing the data for model usage.

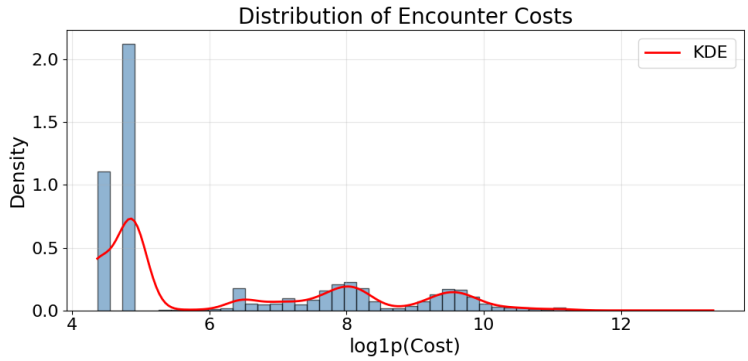

Figure 2: Distribution of log transformed encounter cost

5 Methods

To estimate hospital encounter costs, we implemented and compared two tree-based machine learning models: Extreme Gradient Boosting (XGBoost) and Random Forest. These models were selected for their strong performance on structured healthcare data, particularly in the presence of non-linear relationships and high-dimensional feature spaces.

For the XGBoost model, the training data were used to fit the model, and hyperparameters were tuned using Optuna with Bayesian optimization. The tuning process explored key parameters, including tree depth, learning rate, subsampling ratios, and regularization terms. Early stopping was implemented to prevent overfitting, and model performance was evaluated on the validation set using mean absolute error (MAE). After selecting the optimal hyperparameters, the model was retrained on the full training dataset and evaluated on the validation set. Performance was assessed using standard regression metrics, including mean absolute error (MAE) and the coefficient of determination (R^2), computed on the log-transformed encounter cost. To improve interpretability, these metrics were also calculated on the original dollar scale, including MAE, R^2, and root mean squared error (RMSE).

For the Random Forest model, the training data were similarly used to fit the model, and hyperparameters were tuned using randomized 3-fold cross-validation. The tuning process included parameters such as the number of trees, tree depth, minimum samples per split, minimum samples per leaf, and feature subsampling. The optimal hyperparameters were selected based on minimizing validation MAE. The final model was then retrained on the training data and evaluated on the validation set using the same performance metrics described above.

Table 1: Hyperparameters tuned for XGBoost and Random Forest models

XGBoost	**Random Forest**
▷ `max_depth`	▷ `n_estimators`
▷ `learning_rate`	▷ `max_depth`
▷ `subsample`	▷ `min_samples_split`
▷ `colsample_bytree`	▷ `min_samples_leaf`
▷ `min_child_weight`	▷ `max_features`
▷ `reg_alpha`	
▷ `reg_lambda`	
▷ `n_estimators`	

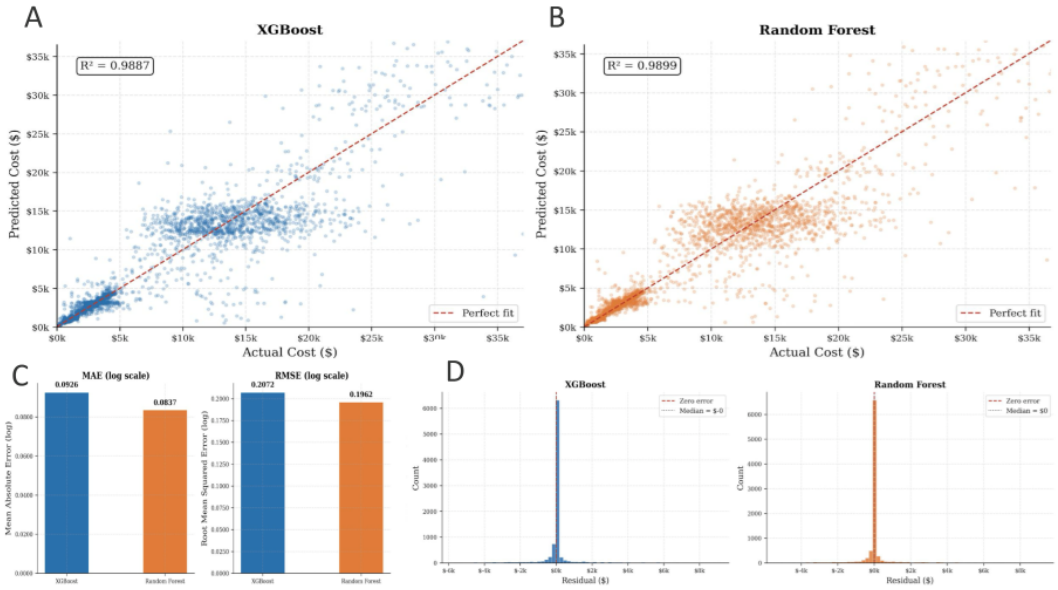

Figure 3: (A)Actual vs. predicted encounter cost for the XGBoost Model on the test set. (B)Actual vs. predicted encounter cost for RF Model on the test set. (C)MAE and RMSE comparison on the log scale. (D) Residual distributions centered near $0 for both models, confirming no systematic bias.

6 Results

Both models achieved strong predictive performance on the held-out test set. Table 2 summarizes key metrics across both evaluation scales.

Table 2: Model performance on the test set

Model	**MAE (log)**	**RMSE (log)**	**R^2 (log)**	**MAE ($)**	**RMSE ($)**
XGBoost	0.0926	0.2072	0.9887	$735	$5,880
Random Forest	0.0837	0.1962	0.9899	$714	$6,546

Both models explained over 98% of the variance in log-transformed encounter cost. Random Forest achieved a slightly lower MAE (714*vs.* 735), while XGBoost produced a lower RMSE ($5,880$*vs.* 6,546), indicating Random Forest is marginally more accurate on average but more sensitive to high-cost outliers. This is crucial for a model to handle in the medical space, since there are always exceptions to the rule. Both models generalized well, with validation and test metrics closely aligned.

Figures A through D illustrate model behavior on the test set. The actual vs. predicted plots (A) show tight clustering along the perfect-fit diagonal at typical cost ranges, with scattering increasing at higher costs where the data are sparse. Residual distributions (D) are symmetric and centered near$0 for both models, which helps confirm the absence of systematic bias. Feature importance analysis identified `procedure_count`, `proc_obstetrics`, and `ENCOUNTER_TIME_MIN` as the strongest predictors of encounter cost, with encounter class

and patient age contributing a moderate signal. Individual condition indicators were less predictive than procedural features, which suggests that what is done during an encounter drives cost more than the underlying diagnosis.

7 Conclusion

This study demonstrates that, while predicting healthcare costs remains a complex challenge due to the wide variability in patient characteristics, data-driven methods can address this problem. Both XGBoost and Random Forest achieved strong predictive performance, explaining over 98% of the variance while maintaining consistent generalization across both the validation and test sets. While Random Forest had better average accuracy, XGBoost was more responsive to high-cost cases, which can often be more volatile or important. Residual analysis confirmed the absence of systematic bias; however, because this data is synthetically generated, it should be interpreted with caution, as real-world scenarios may exhibit additional sources of variability or bias. Feature importance results further indicated that procedural and encounter-related factors are more influential than underlying conditions when determining cost. In conclusion, the developed models provide a tool that enables individuals and organizations to better understand patient cost obligations, thereby improving cost transparency.

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